VibeSync

Project Team

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Chapter 1

Introduction

VibeSync is an AI tool that provides video content analysis for recommending culturally and emotionally appropriate background music. Editors often encounter difficulties in identifying suitable music, especially when their exposure to a specific culture is limited. The aim of this project is to bridge the gap between Pakistani video content and culturally appropriate background music, making the editing process smoother and more efficient.[1]

1.1 Problem Statement

Most editors in the West struggle to match Pakistani cultural elements with the appropriate type of music because they are unfamiliar with the diverse range of music genres practiced in Pakistan. Manual searching for such matches is time-consuming and inefficient, and there is always the risk of assigning inappropriate music to the content. **VibeSync** automates the recommendation of culturally relevant music, resulting in an improved and culturally authentic final product.

1.2 Motivation

A tool must be developed to assist in expressing various cultures as media content becomes increasingly globalized. Video editors working with Pakistani content face the challenge of selecting music that fits the cultural context and emotional tone. **VibeSync** is designed to simplify this process, reducing the time and effort required while providing better cultural accuracy, ultimately enhancing the efficiency of the editor and the experience for the viewer.

1.3 Problem Solution

The proposed solution is an AI-driven software application, **VibeSync**, which automatically recommends culturally and emotionally suitable background music for videos by analyzing features like cultural context, emotional tone, and motion. It assists video editors, especially those unfamiliar with Pakistani cultural elements, by simplifying the music selection process, improving both editing efficiency and cultural authenticity. Using advanced video analysis and a curated music dataset, **VibeSync** provides real-time, culturally relevant music recommendations, reducing manual effort and ensuring seamless integration into existing editing workflows, allowing editors to focus on creative tasks.

1.3.1 Objectives

The key objectives of the **VibeSync** software are:

1. To analyze videos for emotional tone, motion, and scene changes.
2. To detect the cultural context of video content using AI models.
3. To recommend background music that aligns with both cultural and emotional aspects of the video.
4. To create a curated database of culturally relevant Pakistani music for use in the recommendation process.
5. To enhance the cultural and emotional authenticity of video content.
6. To reduce the time and effort required by editors in selecting background music.
7. To improve the global representation of Pakistani culture through accurate audio-visual alignment.

These objectives are designed to address the challenges listed in the problem statement and ensure that **VibeSync** provides an effective and efficient solution for video editors.

1.4 Stakeholders

The success of *VibeSync* depends on various stakeholders who play key roles in its development, deployment, and use. The primary stakeholders involved in the project are:

- **Project Team**

  - **Azan Shahzad:** Responsible for video analysis and cultural identification.
  - **Musharib Nadeem:** Focuses on developing the music recommendation system and integrating all components.

- **Supervisor**
  - **Dr. Arshad Islam:** Provides guidance, feedback, and academic support throughout the project lifecycle.

- **Video Editors/Content Creators**
  - Professionals who will use *VibeSync* to automate the selection of culturally relevant and emotionally resonant background music for their video content.

- **End Users/Viewers**
  - Audiences who will experience improved cultural representation and emotionally resonant content through the enhanced background music recommendations in videos.

Chapter 2

Project Description

This chapter provides a comprehensive description of the proposed project, **VibeSync**, including the scope, core functionalities, and modular structure of the solution.

2.1 Scope

The scope of **VibeSync** involves developing an AI-powered video analysis tool that recommends culturally and emotionally suitable background music for video content. The system's core functionality includes analyzing video features such as motion, emotional tone, scene changes, and cultural context, with a primary focus on Pakistani cultural elements. By automating the music recommendation process, **VibeSync** reduces the time and effort involved in manual selection, ensuring that the recommended music aligns with both the cultural and emotional tone of the video, enhancing the overall viewing experience. Seamlessly integrated into existing video editing workflows, the tool will utilize a curated dataset of Pakistani music genres to provide culturally relevant audio recommendations. Additionally, the system is designed for scalability, allowing it to accommodate various types of video content across multiple genres and formats. Future enhancements may include expanding the cultural database to cover global music genres and improving AI models for more complex video analysis.

2.2 Modules

The proposed project, **VibeSync**, consists of several key modules that address different aspects of video analysis and music recommendation. These modules are designed to work together to provide a seamless experience for the user.

Figure 2.1: Workflow Diagram

2.2.1 Client Web App

1. **Video Upload Module**: Allows users to upload videos for analysis.
2. **Video Analysis Module**: Analyzes videos for features such as motion, scene changes, and emotional tone.
3. **Music Recommendation Module**: Recommends suitable background music based on the analysis of the video.[2]
4. **Integration Module**: Seamlessly integrates the recommended music with the video, allowing users to download or export the final video with the background music.

2.3 Tools and Technologies

The development of **VibeSync** leverages various tools and technologies to analyze video content and recommend culturally relevant background music. The following are the key tools and technologies required for this project:

- **ResNet & BERT**: These models are used for evaluating the cultural context in videos by analyzing visual and audio cues.[3]
- **CLIP**: A model used to detect emotional content and semantic features within video frames.

- **OpenCV**: A computer vision library used to calculate motion between video frames, helping identify rhythm and dynamic elements.
- **SceneDetect**: This tool detects scene changes in videos, segmenting the video into meaningful parts for better analysis.
- **Python**: The core programming language used to implement the AI models, integrate video analysis, and manage the recommendation system.
- **Music Dataset**: A custom dataset containing culturally relevant Pakistani music genres, enabling culturally and contextually accurate music recommendations.
- **AI Models**: Various AI techniques to identify motion, emotional tone, and scenes, which are crucial to matching videos with suitable background music.

2.4 Work Division

The following table outlines the work division among the project team members, assigning each team member specific responsibilities based on the modules and features of the project.

Table 2.1: Table 1: Work Division

Name	**Registration**	**Responsibility / Module / Feature**
Azan Shahzad	22I-2068	Video analysis (cultural identification, emotion detection)
Musharib Nadeem	20I-1764	Music recommendation system, integration with video analysis
Azan Shahzad	22I-2068	Motion detection, Scene change detection
Musharib Nadeem	20I-1764	Dataset creation, Music recommendation integration
Azan Shahzad	22I-2068	API's Backend
Musharib Nadeem	20I-1764	Web Development

2.5 Timeline

The timeline for the project is structured across several iterations, each focused on specific tasks and modules. The following table presents the planned timeline, mapping tasks to their respective time frames.

Table 2.2: Table 2: Project Timeline

Iteration #	**Time frame**	**Tasks / Modules**
01	Sept-Nov	Web development
02	Nov-Jan	Video Analysis and Model Training
03	Feb-Apr	Music recommendation system, Dataset Creation
04	Apr-June	Deployment, Testing, and Feedback

Chapter 3

System Overview

VibeSync is a tool that uses AI to help video editors pick the right background music for videos, especially ones with Pakistani culture. It analyzes video features like movement, emotions, and cultural details to suggest appropriate music, making the editing process easier.

The project helps solve the problem of choosing music that matches visuals, especially for editors unfamiliar with Pakistani culture. **VibeSync** uses AI to ensure the music fits the culture and emotions portrayed in the video.

VibeSync consists of several components: video upload, video analysis, music recommendation, and integration. The AI considers the cultural and emotional aspects of the video, and the recommendation system assists users in selecting the right music. The system is designed to scale easily, get updates, and expand to include other cultural contexts in the future.

3.1 Architectural Design

The architecture of **VibeSync** is modular, where each component performs a specific task while interacting efficiently with others. The system uses a multi-tiered client-server architecture, making it easier to manage and scale.

3.1.1 System Overview

VibeSync is built using a Multi-Tiered Client-Server architecture, which separates the system into different layers: user interaction, application processing, and data management. This design allows each part to focus on its specific responsibilities, making the system easier to manage and expand.

3.1.2 Modular Program Structure

1. **Client Web App**: The client web app is the user interface (UI) that video editors use to interact with the system. It lets users upload videos, set preferences, and view recommendations.

2. **Video Analysis Module**: This module automatically analyzes the uploaded videos for key features like emotion, scene changes, and movement using AI tools like CLIP and OpenCV.

3. **Cultural Context Detection Module**: This module detects the cultural context of the video using AI models like ResNet and BERT by analyzing visual and audio cues.

4. **Music Recommendation Module**: Based on the video analysis and cultural context detection, this module recommends suitable background music from a curated music library.

5. **Integration Module**: Once the music is chosen, this module integrates the selected music into the video. The final edited video can then be downloaded or exported.

6. **Database and Backend**: This module handles data storage, including video files, music, and analysis results, ensuring smooth system performance and data management.

3.1.3 Inter-Module Relationships

1. The **Client Web App** sends the uploaded video to the **Video Analysis Module** for processing.

2. The **Video Analysis Module** collaborates with the **Cultural Context Detection Module** to detect cultural elements in the video.

3. Both the **Video Analysis** and **Cultural Context Detection Modules** provide data to the **Music Recommendation Module**, which determines the best background music.

4. The **Integration Module** combines the recommended music with the original video and returns the final version to the **Client Web App**.

5. The **Database** stores video content, analysis data, and music to maintain system consistency.

3.2 Design Models

VibeSync employs various design models to illustrate how the system is structured and functions. These models ensure that all components work seamlessly together.

3.2.1 Object-Oriented Development Models

1. **Activity Diagram**: Shows the steps in the workflow, such as uploading videos, analyzing them, recommending music, and integrating it into the video.
2. **Class Diagram**: Illustrates the system's classes, such as **VideoAnalysis**, **CulturalContext**, **MusicRecommendation**, and **Integration**, along with their attributes and relationships.
3. **Class-Level Sequence Diagram**: Depicts how the system's components interact over time. For instance, it demonstrates how the **Client Web App** communicates with the **Video Analysis**, **Cultural Context Detection**, and **Music Recommendation** modules.
4. **State Transition Diagram**: Illustrates the system's state changes, such as transitioning from "Video Uploaded" to "Music Integrated."

3.3 Data Design

In **VibeSync**, data design explains how information is organized into data structures. This section details how data is stored, processed, and organized to ensure smooth system performance.

3.3.1 Data Structures and Storage

1. **Video Data**: Uploaded videos are stored as binary files. Information like duration, size, format, and upload date is also saved to assist in analysis.
2. **Video Analysis Data**: This data, generated by the **Video Analysis Module**, includes emotions, scene changes, and movement, stored as structured records in the database.
3. **Cultural Context Data**: Detected by the **Cultural Context Detection Module**, this data is saved in **JSON format**, allowing flexibility in storing various types of information like cultural identifiers and emotional tags.

4. **Music Recommendation Data**: The **Music Recommendation Module** chooses appropriate background music based on the analysis. The music library is organized in a **relational database**, linking recommended music to the respective video.

5. **User Data**: User profiles, preferences, and uploaded video history are stored in a **relational database** to offer personalized recommendations and manage user activity.

3.3.2 Data Processing and Organization

- **Database Management System (DBMS)**: A relational database stores user data, the music library, and video information for fast searching and retrieval.
- **Music Library**: The music library contains a collection of Pakistani songs, organized by attributes like genre, emotion, and cultural significance, stored in a **SQL database**.
- **Backend Data Handling**: Backend services manage data processing, like running analysis models and storing results. The backend also ensures smooth communication between different system modules.

3.3.3 Data Storage Items

- **Video Files Storage**: Videos are stored in a scalable cloud-based system.
- **Relational Database**: Structured data, such as user profiles, music details, and analysis results, is stored in a relational database.
- **JSON Data**: Some information, such as cultural context and analysis details, is stored in **JSON format** for flexibility.

This data design helps **VibeSync** efficiently organize, process, and manage its data, ensuring accurate music recommendations and seamless integration into videos.

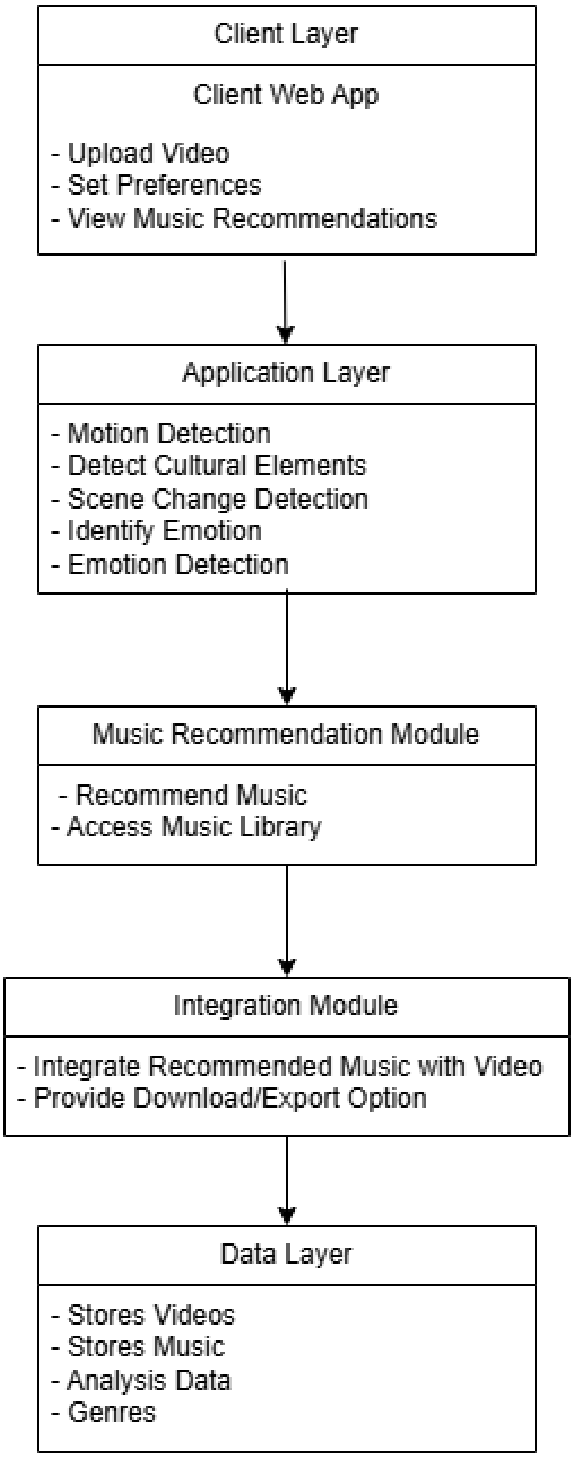

Figure 3.1: Architecture Diagram

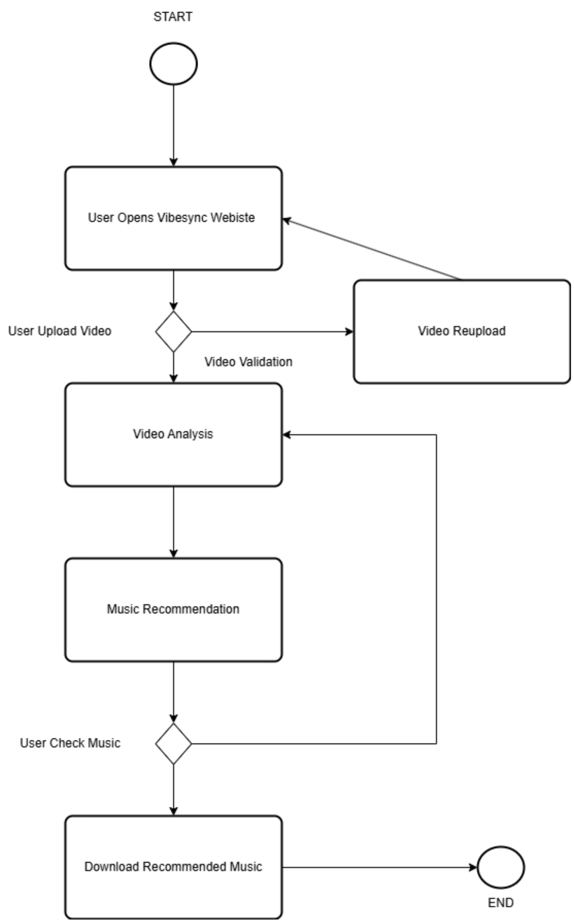

Figure 3.2: Activity Diagram

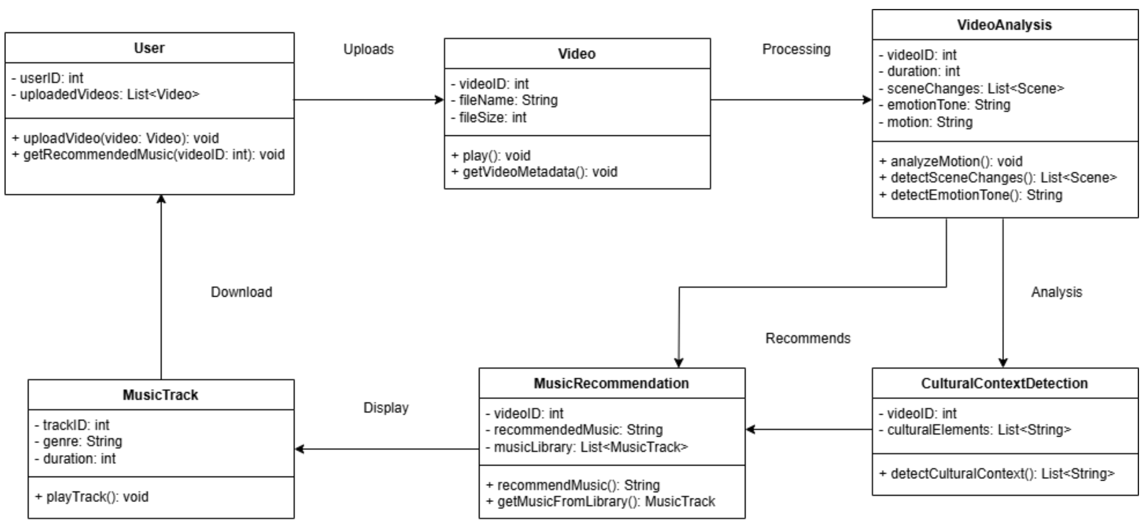

Figure 3.3: Class Diagram

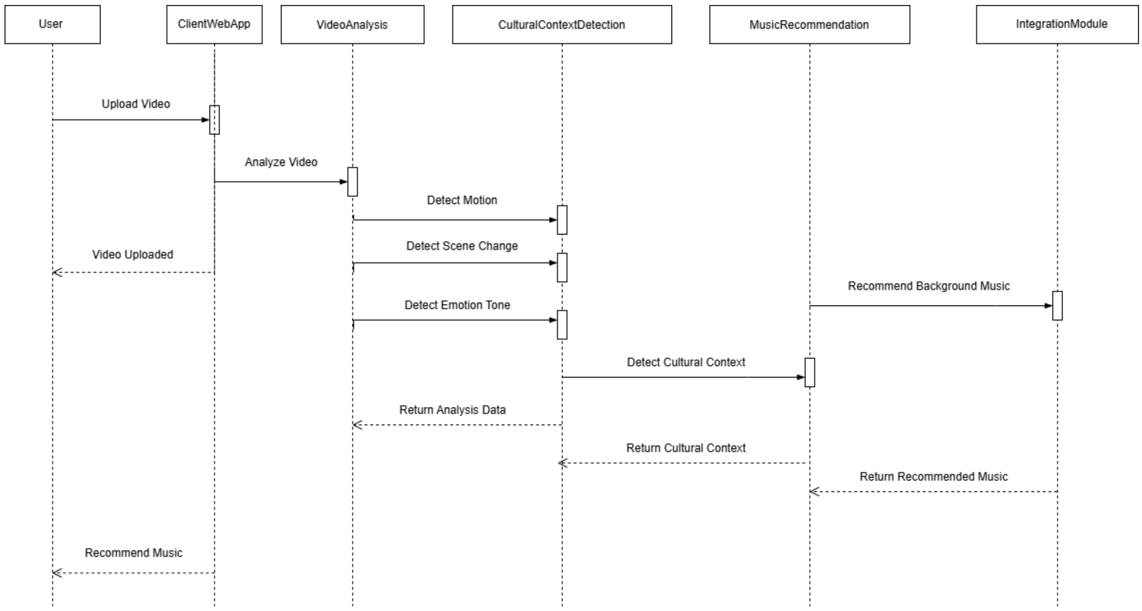

Figure 3.4: Class-Level Sequence Diagram

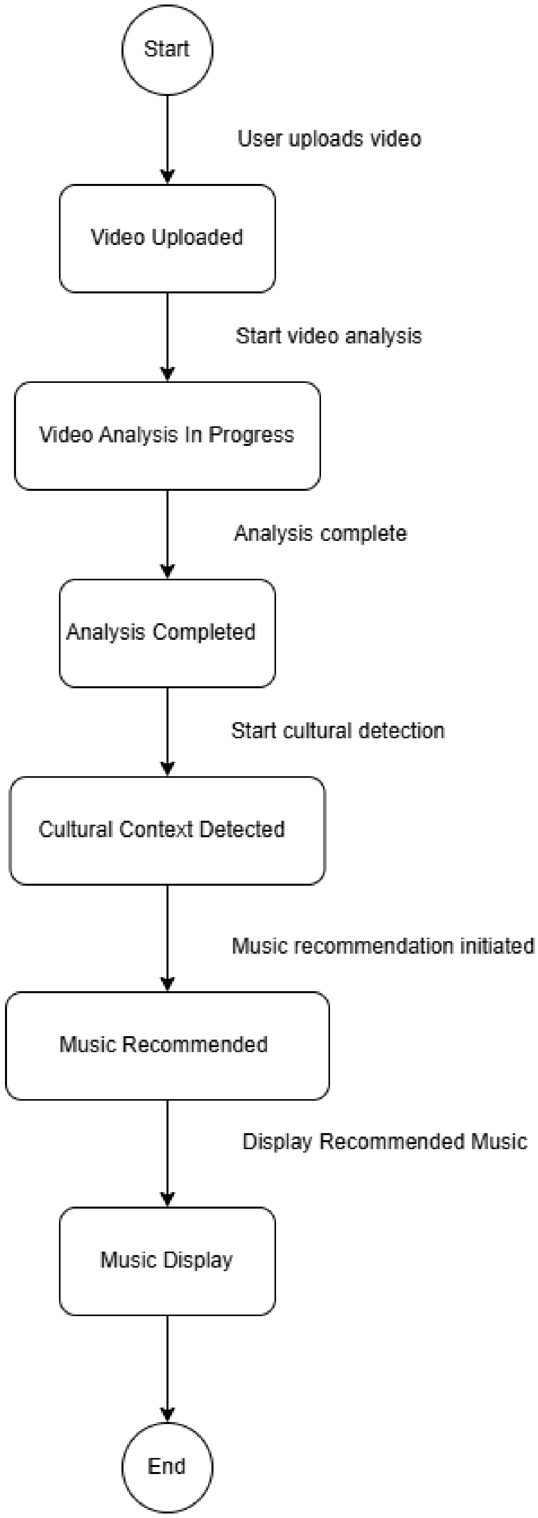

Figure 3.5: State Transition Diagram

Chapter 4

Conclusions and Future Work

4.1 Conclusions

In conclusion, **VibeSync** successfully addresses the challenge of recommending culturally and emotionally suitable background music for video content, with a specific focus on Pakistani culture. Through advanced video analysis techniques and AI-driven music recommendations, the system improves the efficiency of video editing by automating the process of selecting culturally relevant music. By analyzing motion, emotional tone, and cultural context, **VibeSync** ensures that the recommended music not only aligns with the visual content but also enhances the overall viewing experience.

The project has demonstrated the potential of using artificial intelligence to bridge cultural gaps in media production, providing a tool that is both practical for editors and beneficial for the accurate representation of diverse cultures.

4.2 Future Work

Future work for **VibeSync** includes expanding its scope to cover global cultural contexts beyond Pakistan. The system can be adapted to analyze and recommend music for video content from various regions, such as China, Japan, Russia, and many others. By integrating more diverse datasets representing different musical traditions and cultural cues, **VibeSync** can evolve into a globally applicable tool for video editors, enabling culturally accurate music recommendations for content from around the world. Additionally, further enhancements in AI models and real-time recommendations will be explored to improve the user experience and broaden the system's capabilities.

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